

Yuda Risma Wahyudi

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Experience

Senior Automation Test Engineer

PEGATRON

September 2022 - present

Batam, Indonesia

- Developed automated test programs for the production line FATP (Final Assembly Test and Pack), enhanced testing algorithms and optimize program structures, resulting in a 20% reduction in testing cycle time.
- Led the development of reusable libraries for communication between testing software and instruments, improving maintainability and efficiency using .NET for Windows applications and PHP for web platforms.
- Deployed software debugging tools used by 100+ engineers across factories in Taipei, Batam, Suzhou, and Vietnam, significantly enhancing troubleshooting workflows and reducing downtime by 30%.
- Conducted deep research on 5G OTA testing methodologies, evaluated antenna performance, RF characteristics, signal quality, and compliance with 3GPP standards. Designed test flows for various 5G projects (dongle, camera, and FWA), setting up environments, and defining specifications with customer.
- Directed troubleshooting and maintenance activities across production lines, guiding engineers to resolve issues and delivering customer-driven solutions

Research Assistant

Assistive Robotic Laboratory, National Taipei University of Technology

August 2021 – July 2022

Taipei, Taiwan

- Researched and developed a robotic teaching method using haptic devices to transfer human motion into robotic control, significantly reducing robot programming time for complex tasks such as polishing, resulting in 2 US patents and 3 journals publication.
- Contributed to the industry-funded research projects on automated industrial systems, collaborating with cross-functional teams to deliver practical solutions.
- Developed a real-time robotic automation inspection system (AOI) using deep learning and computer vision algorithms for efficient and robust defect detection on product surfaces.
- Reviewed relevant literature, implemented state-of-art control algorithms (including robot kinematics, robot navigation, control theory, and motion planning), contributed to the invention of new methods and approaches.
- Developed robot APIs and communication protocols enabling seamless integration between computers and various industrial robots such as Fanuc, Yaskawa, and Denso.

Head of Robotic Team

Robotic Laboratory, State University of Malang

2016 - 2018

Malang, Indonesia

- Managed a cross functional team to design and build a hexapod robot, maintain project timelines, resource allocation, and technical development.
- Led the design and implementation of robotic control and navigation systems, developing all algorithms in C++ on STM32 controllers to deliver efficient, reliable, and precise robot motion.
- Published a paper on navigation control of a hexapod robot.
- Compete in the National Indonesia Fire-Fighting Robot Contest 2017, achieving best 5th finalist position among 101 teams.

Education

National Taipei University of Technology

Master of Science in Automation Technology • GPA 4.00

Taipei, Taiwan

September 2019 – August 2021

State University of Malang

Bachelor of Engineering in Electrical Engineering • GPA 3.70

Malang, Indonesia

July 2014 – December 2018

Skills

Programming

: C++, C#, Python, PHP, Matlab

Development Tools

: Visual Studio, VS Code, ROS, PyCharm, Git, Laravel, Vite

Languages

: English (proficient), Indonesian (native), Chinese (conversational)